

BAV — Exam Question Bank

Business Analysis & Valuation

45 solved multiple-choice questions

Forty questions on the taught course, plus five on the guest-lecture session. Every question carries four options, a full worked solution and the answer. Numbers are taken from the Ambuja and UltraTech valuations built for the project, so the arithmetic is checkable against the models.

What is in here

A	The five consistencies	Q1–8
B	DCF variants and why they must agree	Q9–17
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How to use it. Cover the working with a sheet of paper, answer the four options, then check. The starred ★ **Remember** lines are the traps that separate marks — most of them are sign errors, wrong-rate errors or double counting, not hard arithmetic.

Note on Section F: the guest-lecture questions are written on M&A, synergy valuation and exchange ratios, which is the module the course carries separately from the core valuation syllabus. If the guest session covered a different topic, these five should be swapped.

SECTION A The five consistencies

Mohanty's central claim: a valuation is correct when it preserves five consistencies — assumptions, risk, asset, liability, multiplier. Maintain all five and every DCF variant returns the same equity value.

1

Five consistencies — risk

A student discounts free cash flow to equity at the WACC. Which consistency has been broken?

- (a) Assumptions consistency
- (b) Risk consistency
- (c) Asset consistency
- (d) Multiplier consistency

Working. Risk consistency requires the cash flow and the discount rate to belong to the same claimholder. FCFE is an equity flow, so it must be discounted at the cost of equity K_e . WACC is a firm-level rate and belongs with FCFF or CCF.

Firm flows $\rightarrow$ WACC or ρ | Equity flows $\rightarrow K_e$

Since $WACC < K_e$ whenever there is debt, this error *overstates* equity value.

Ans: (b) Risk consistency

★ **Remember.** The classic pair is FCFE at WACC and FCFF at K_e . Check the label on the cash flow before you check the arithmetic.

2

Five consistencies — assumptions

A forecast grows sales at 15% a year while holding capital expenditure and working capital flat in rupee terms. Which consistency fails?

- (a) Assumptions consistency
- (b) Liability consistency
- (c) Risk consistency
- (d) Multiplier consistency

Working. Growth requires reinvestment. Selling more cement needs more plant and more working capital; a model that grows revenue without growing the capital base is manufacturing free cash flow out of nothing.

$g = \text{RONIC} \times \text{reinvestment rate}$

Rearranged, reinvestment rate = g/RONIC . If reinvestment is zero, g must be zero unless RONIC is infinite.

Ans: (a) Assumptions consistency

★ **Remember.** Every growth assumption has a matching investment assumption. If you cannot show where the money went, the growth is not real.

3

Five consistencies — asset

A model excludes interest income from NOPAT because cash is treated as non-operating, but forgets to add the cash balance back in the equity bridge. What is the effect?

- (a) Equity value is unchanged
- (b) Equity value is overstated
- (c) Equity value is understated
- (d) Only the WACC is affected

Working. Asset consistency: every asset is counted exactly *once*. Two valid treatments exist —

(i) income *in* NOPAT → asset *not* added back

(ii) income *excluded* → asset added back at the bridge

Excluding the income and also omitting the add-back counts the cash zero times. The company owns the cash but the valuation gives no credit for it, so value is understated by the cash balance.

Ans: (c) Equity value is understated

★ **Remember.** Asset errors come in both directions. Adding back land while its rent still sits in operating income counts it twice; this question is the mirror image.

4

Five consistencies — liability

A deferred sales-tax liability is included as a source of funds in the WACC weights but is not subtracted in the bridge from enterprise value to equity. Which consistency fails, and in which direction?

(a) Liability; equity overstated

(b) Liability; equity understated

(c) Asset; equity overstated

(d) Risk; no directional effect

Working. Liability consistency: whatever is included in the cost of capital as a source of funds *must* be subtracted from enterprise value to reach equity. Including the liability in the weights but not in the bridge means the claim finances the assets but never gets repaid out of the value.

Equity = EV + non-operating assets — every prior claim

The omitted deduction inflates the residual left for shareholders.

Ans: (a) Liability; equity overstated

★ **Remember.** Non-interest-bearing items are either inside net working capital or treated as debt-like — never both, and never neither.

5

Five consistencies — multiplier

A model assumes terminal growth of 10% and produces an implied terminal EV/EBITDA of 4x. What does this indicate?

(a) Nothing; multiples and DCF are unrelated

(b) Multiplier inconsistency

(c) The WACC must be too low

(d) The forecast horizon is too long

Working. A multiple is the *output* of value divided by an accounting indicator, so the multiple your model implies must be arithmetically consistent with the growth, return and discount rate the model itself uses.

$$\text{Justified } \frac{EV}{EBITDA} = \frac{(NOPAT/EBITDA)(1 - g/ROIC)}{WACC - g}$$

High terminal growth with a very low exit multiple is internally contradictory: a company growing at 10% forever would command a high multiple, not 4x.

Ans: (b) Multiplier inconsistency

6

Method equivalence

All five DCF methods in a model return exactly the same value per share. What does this prove?

- (a) The assumptions are correct
- (b) The valuation is accurate
- (c) The model is internally consistent, and nothing more
- (d) The market price must be wrong

Working. The five methods are one model rearranged. Agreement demonstrates that no consistency has been broken in the *construction* of the model. It says nothing whatever about whether the inputs are right.

Agreement $\Rightarrow$ internal consistency **but** internal consistency $\nRightarrow$ correct value

A model with a badly wrong growth assumption will still reconcile across all five methods, perfectly and pointlessly.

Ans: (c) The model is internally consistent, and nothing more

★ **Remember.** This is a favourite one-mark question. Never write that reconciliation proves the valuation is right.

7

Five consistencies — liability

Operating leases are capitalised and the lease liability is deducted in the equity bridge, but rent continues to be expensed inside the free cash flow. What has happened?

- (a) Correct treatment
- (b) The lease is counted twice as a cost
- (c) The lease is not counted at all
- (d) Only the beta is affected

Working. Capitalising the lease means treating it as debt: the liability is deducted at the bridge and the rent should be *removed* from operating expense (replaced by depreciation and interest, both of which sit below FCFF). Leaving rent in the cash flow *and* deducting the liability charges the company for the same obligation twice — once through lower FCFF and once through a smaller bridge.

Ans: (b) The lease is counted twice as a cost

8

Terminal reinvestment

A model charges terminal reinvestment as g times invested capital *including* acquisition goodwill. What does this implicitly assume?

- (a) That the company must keep buying goodwill forever merely to grow (b) That goodwill depreciates
(c) That the tax shield is perpetual (d) Nothing — it is the standard treatment

Working. Reinvestment in the terminal year funds the assets that produce the extra output. Organic growth needs more plant and more working capital; it does not need the company to keep making acquisitions.

$$\text{Terminal reinvestment} = g \times \text{operating invested capital}$$

Goodwill still belongs *inside* invested capital when measuring the return shareholders earned on what they paid — but not inside the reinvestment charge. In the Ambuja model, getting this wrong drove the equity value negative.

Ans: (a) That the company must keep buying goodwill forever merely to grow

★ **Remember.** Goodwill: in the ROIC denominator, out of terminal reinvestment.

SECTION B DCF variants and why they must agree

The OLP Limited test vector runs through Q9–Q12. Inputs: net operating income $X = \text{Rs } 100\text{m}$ in perpetuity, tax $t = 20\%$, debt $D = \text{Rs } 100\text{m}$ at $K_d = 10\%$, cost of equity $K_e = 20\%$, no non-operating assets, zero growth.

9

OLP — FCF at WACC

Using the OLP inputs, what is the enterprise value under the enterprise FCF method?

- (a) Rs 400m (b) Rs 440m
(c) Rs 460m (d) Rs 500m

Working. Step 1 — the cash flow. With zero growth there is no reinvestment, so $\text{FCF} = X(1 - t) = 100(0.8) = \text{Rs } 80\text{m}$.

Step 2 — equity value, needed for the weights. $\text{FCFE} = (X - K_d D)(1 - t) = (100 - 10)(0.8) = 72$, so $E = 72/0.20 = \text{Rs } 360\text{m}$ and $V = 360 + 100 = 460$.

Step 3 — the WACC.

$$\text{WACC} = \frac{E}{V} K_e + \frac{D}{V} K_d (1 - t) = \frac{360}{460} (0.20) + \frac{100}{460} (0.10)(0.8) = 17.391\%$$

Step 4 — $EV = 80/0.17391 = \text{Rs } 460\text{m}$.

Ans: (c) Rs 460m

10

OLP — FCFE at K_e

Using the same inputs, what is the equity value under the FCFE method?

- (a) Rs 360m (b) Rs 400m
(c) Rs 460m (d) Rs 288m

Working. Equity cash flow is what is left after paying interest and tax, with zero growth so no net new borrowing.

$$FCFE = (X - K_d D)(1 - t) = (100 - 10)(0.8) = \text{Rs } 72\text{m}$$

Discount at the *equity* rate: $E = 72/0.20 = \text{Rs } 360\text{m}$.

Cross-check: $EV = E + D = 360 + 100 = 460$, matching Q9 exactly. That equality is the whole point of the exercise.

Ans: (a) Rs 360m

11

OLP — CCF at pre-tax WACC

What is the capital cash flow, and the rate at which it is discounted?

(a) 80 at 17.391%

(b) 82 at 17.826%

(c) 72 at 20.000%

(d) 82 at 17.391%

Working. Capital cash flow adds the interest tax shield back into the flow, so the shield is no longer inside the discount rate — which means the rate must be the *pre-tax* WACC.

$$CCF = FCF + t K_d D = 80 + (0.20)(0.10)(100) = \text{Rs } 82\text{m}$$

$$\text{pre-tax WACC} = \frac{E}{V} K_e + \frac{D}{V} K_d = \frac{360}{460}(0.20) + \frac{100}{460}(0.10) = 17.826\%$$

$EV = 82/0.17826 = \text{Rs } 460\text{m}$. Same answer again.

Ans: (b) 82 at 17.826%

★ **Remember.** If you add the shield to the numerator you must take it out of the denominator. Doing both, or neither, is the error.

12

OLP — APV

Under APV with debt fixed in rupee terms, what are the unlevered cost of capital and the value of the tax shield?

(a) $K_u = 20.000\%$, VTS = Rs 10m

(b) $K_u = 18.182\%$, VTS = Rs 20m

(c) $K_u = 17.391\%$, VTS = Rs 20m

(d) $K_u = 18.182\%$, VTS = Rs 80m

Working. Unlever using the MM (1963) relation, which is the fixed-debt case:

$$K_e = K_u + (K_u - K_d) \frac{D}{E} (1 - t)$$

$0.20 = K_u + (K_u - 0.10) \frac{100}{360} (0.8)$. Solving, $K_u = 18.182\%$.

With debt fixed and zero growth the shield is a perpetuity discounted at K_d , which collapses to

$$\text{VTS} = \frac{t K_d D}{K_d} = t D = 0.20 \times 100 = \text{Rs } 20\text{m}$$

$\text{APV} = 80/0.18182 + 20 = 440 + 20 = \text{Rs } 460\text{m}$.

Ans: (b) $K_u = 18.182\%$, VTS = Rs 20m

13

FCFF to FCFE bridge

Which identity converts free cash flow to the firm into free cash flow to equity?

- (a) $FCFE = FCFF - \text{interest} + \text{tax}$ (b) $FCFE = FCFF - \text{interest}(1 - t) + \text{net new borrowing}$
 (c) $FCFE = FCFF + \text{interest}(1 - t) - \text{net new borrowing}$ (d) $FCFE = FCFF - \text{dividends}$

Working. FCFF is the cash available to *all* providers of capital. To get to the equity holders you must pay the lenders and then add back anything the lenders newly advanced.

$$FCFE = FCFF - I(1 - t) + \Delta D$$

Interest is taken after tax because it is tax-deductible. Net new borrowing ΔD is a genuine cash inflow to equity.

Ans: (b) $FCFE = FCFF - \text{interest}(1 - t) + \text{net new borrowing}$

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Net new borrowing

A firm grows at g forever and maintains a constant debt ratio. What is net new borrowing each year?

- (a) Zero (b) $g \times D$
 (c) $K_d \times D$ (d) $t \times K_d \times D$

Working. A constant debt *ratio* with a firm growing at g means the rupee amount of debt must also grow at g ; otherwise the ratio falls every year.

$$\Delta D = g \times D$$

This is the most common error in the course: students discount a growing FCFE at a constant K_e while implicitly holding debt fixed. Equity flows then grow faster than g , the debt ratio drifts, and the constant- K_e assumption collapses.

Ans: (b) $g \times D$

★ **Remember.** If you hold WACC constant you have assumed a constant D/E, which *requires* debt to grow at g . Put the net borrowing line in the model and show it.

15

Choice of method

A company is deleveraging on a announced schedule over five years. Which valuation method is most appropriate?

- (a) FCFF at a single constant WACC (b) APV
(c) Relative valuation only (d) Dividend discount model

Working. A single constant WACC embeds a constant capital structure. If leverage is changing on a known path, the WACC changes every year and one average rate is simply wrong.

APV separates the two effects: value the assets unlevered at K_u , then add the present value of the year-by-year tax shields the actual debt schedule produces.

$$APV = V_u + PV(\text{tax shields})$$

Ans: (b) APV

★ **Remember.** APV is the LBO and deleveraging method precisely because the financing side is modelled explicitly rather than buried in a rate.

16

EVA reconciliation

Enterprise value under EVA equals opening invested capital plus the present value of future EVA. Which two conditions are necessary for this to tie to the FCFF answer?

- (a) Positive EVA, and a five-year horizon (b) FCFF defined as NOPAT less the change in invested capital, and a true steady-state terminal year
(c) A constant WACC, and zero growth (d) Goodwill excluded from invested capital

Working. EVA is not an independent method; it is the same cash flows re-expressed.

$$EVA_t = NOPAT_t - WACC \times IC_{t-1}$$

$$EV = IC_0 + \sum \frac{EVA_t}{(1 + WACC)^t}$$

The identity holds only if (i) FCFF is defined consistently as $NOPAT - \Delta IC$, so that capital charged equals capital invested, and (ii) the terminal year is a genuine steady state, so the perpetuity of EVA matches the perpetuity of FCFF. Drop either and the two answers diverge.

Ans: (b) FCFF = NOPAT – ΔIC, and a true steady-state terminal year

17

When methods disagree

Your FCFF valuation gives Rs 520 per share and your FCFE valuation gives Rs 480. What should you do?

- (a) Average them to Rs 500 (b) Report the more conservative figure
(c) Find and fix the inconsistency (d) Report both as a range

Working. The methods are algebraically identical when the five consistencies hold. A gap is not a modelling choice or a legitimate range — it is proof of an error.

The three usual culprits, in order of frequency: debt not growing at g where a constant WACC was assumed; the tax shield double-counted (in the rate *and* in the flow); cash or investments treated differently in the flows and in the bridge.

Ans: (c) Find and fix the inconsistency

★ **Remember.** Never average. Averaging two answers that must be equal hides the error and throws away the diagnostic.

SECTION C Cost of capital, relevering and circularity

Nine questions on the inputs that decide the discount rate — and on the two places where students most often break risk consistency without noticing.

18

Risk-free rate

The 10-year G-Sec yield is 6.85% and the India sovereign default spread is 1.87%. The equity risk premium being used already includes a country risk premium. What risk-free rate should be used?

- (a) 6.85% (b) 8.72%
(c) 4.98% (d) 7.08%

Working. The G-Sec is not default-free: India is rated Baa3, so the yield contains compensation for sovereign default risk. If the ERP already carries a country risk premium, using the raw G-Sec counts India risk *twice*.

$$r_f = 6.85\% - 1.87\% = 4.98\%$$

The rule is symmetric: strip the spread from the risk-free rate *or* use an ERP without a country premium. Do one, never both and never neither.

Ans: (c) 4.98%

★ **Remember.** Double-counting country risk is a silent error — the number still looks plausible. State which convention you used.

19

Relevering — Hamada

Unlevered beta is 0.90, market D/E is 6.4681% and the marginal tax rate is 25.168%. What is the levered beta under the fixed-debt (Hamada) formula?

- (a) 0.9436 (b) 0.9582
(c) 0.9729 (d) 0.9000

Working.

$$\beta_L = \beta_U \left[1 + (1 - t) \frac{D}{E} \right]$$

$$= 0.90 \left[1 + (1 - 0.25168)(0.064681) \right]$$

$$= 0.90 \left[1 + (0.74832)(0.064681) \right] = 0.90 \times 1.048403 = \mathbf{0.9436}.$$

Ans: (a) 0.9436

★ **Remember.** The tax term is $(1 - t)$, not $(1 + t)$. A sign slip here is a real error that appeared in one of the project workbooks.

20

Relevering — Harris–Pringle

The same company states a policy of constant debt-to-value with continuous rebalancing, and discounts its tax shield at ρ . What is the correct levered beta?

(a) 0.9436

(b) 0.9582

(c) 0.9000

(d) 0.9729

Working. Under Case 2 — constant D/V , continuously rebalanced — the debt level moves with firm value, so the tax shield carries *asset* risk rather than debt risk. It is discounted at ρ , and the relevering formula carries **no tax term**.

$$\beta_L = \beta_U \left(1 + \frac{D}{E} \right) = 0.90 (1.064681) = \mathbf{0.9582}$$

Using the Hamada formula while discounting the shield at ρ mixes Case 1 and Case 2 and breaks risk consistency.

Ans: (b) 0.9582

★ **Remember.** Match the unlever, the relever and the shield discount rate to *one* debt policy. This is the single most common risk-consistency failure in real models.

21

The ρ identity

WACC is 11.4286%, K_d is 8.34%, the tax rate is 25.168% and D/V is 6.0751%. What is the unlevered cost of capital ρ ?

(a) 11.4286%

(b) 11.5561%

(c) 11.3011%

(d) 11.7641%

Working.

$$\rho = \text{WACC} + K_d t \frac{D}{V}$$

$$= 0.114286 + (0.0834)(0.25168)(0.060751)$$

$$= 0.114286 + 0.001275 = \mathbf{11.5561\%}.$$

Equivalently $\rho = \frac{E}{V} K_e + \frac{D}{V} K_d$ — the unweighted mix with no tax relief, because under CCF and APV the shield has been taken out of the rate.

Ans: (b) 11.5561%

★ **Remember.** This identity is the fastest audit in the whole model. If ρ and WACC do not satisfy it to the last decimal, something is wrong.

22

Debt policy and the shield

Under which debt policy is the interest tax shield discounted at the cost of debt K_d ?

- (a) Constant D/V , continuously rebalanced (b) Debt fixed in rupee terms
(c) Constant D/V , rebalanced annually (d) All three

Working. The discount rate for the shield must match the risk of the shield, which follows the risk of the debt.

Case 1 — debt fixed in Rs: shield is as safe as the debt → discount at K_d

Case 2 — Harris–Pringle: debt moves with value → discount at ρ

Case 3 — Miles–Ezzell: first year at K_d , later years at ρ

Only fixed rupee debt gives a shield whose amount is known in advance and therefore carries debt risk throughout.

Ans: (b) Debt fixed in rupee terms

23

Circularity

WACC needs market-value weights, but the equity value is what you are solving for. What are the two sanctioned solutions?

- (a) Use book-value weights, or ignore the problem (b) Use a target D/E, or iterate to convergence
(c) Use the industry average, or use book values (d) Use the risk-free rate instead of WACC

Working. Both solutions are legitimate and both must be disclosed.

(i) **Target D/E.** Use management's stated or the industry's structure. Appropriate when the firm will migrate to it. Only the *ratio* is needed, never rupee values.

(ii) **Iterative convergence.** Seed with market capitalisation, compute WACC → value → implied equity, feed that back and repeat until (implied – assumed) = 0. In Excel: enable iterative calculation or use Goal Seek.

For a listed company, using market cap is acceptable *if stated* — but note the tension: you are using the very price you claim may be wrong.

Ans: (b) Use a target D/E, or iterate to convergence

★ **Remember.** Never silently use book-value weights. That is an automatic mark lost.

24

Tax rate

A company has migrated to section 115BAA. What is the effective marginal corporate tax rate?

- (a) 22.000% (b) 25.168%
(c) 34.944% (d) 17.160%

Working. Section 115BAA is a 22% base rate, plus a 10% surcharge, plus a 4% health and education cess. The loadings are multiplicative, not additive.

$$22\% \times 1.10 \times 1.04 = \mathbf{25.168\%}$$

MAT does not apply under 115BAA. The old regime works out near 34.944% and section 115BAB (new manufacturing) near 17.16%. Always verify the regime from the tax reconciliation note — never assume.

Ans: (b) 25.168%

25

Cost of debt

Which is the preferred source for the pre-tax cost of debt?

- (a) Interest expense divided by average debt (b) The yield on the company's traded bonds, or the rate disclosed in the borrowings note
(c) The risk-free rate plus 2% (d) The coupon on the oldest outstanding loan

Working. K_d must be a *forward-looking market* rate — what the company would pay to borrow today. Interest expense over average debt is a historical average of legacy borrowings at legacy rates, distorted by drawdowns and repayments during the year.

Use it only as a sanity check. And remember the after-tax conversion for the WACC:

$$K_d^{\text{after-tax}} = K_d(1 - t)$$

Ans: (b) The yield on the company's traded bonds, or the rate disclosed in the borrowings note

26

Beta

A regression beta from 60 monthly observations is 1.15 with an R^2 of 0.27. The bottom-up beta from 469 sector peers is 0.91. Which should you use, and why?

- (a) The regression beta, because it uses the company's own price history (b) The bottom-up beta, because the regression explains little of the variance
(c) The average of the two (d) Whichever gives the higher valuation

Working. An R^2 of 0.27 means the market explains only 27% of the stock's variation — the regression estimate carries a wide standard error and is unstable to the choice of window and frequency.

A bottom-up beta averages the unlevered betas of many firms in the same business, so estimation errors diversify away, and it is then relevered at the subject's own D/E. Where the two diverge materially, prefer bottom-up and *explain* the choice.

Ans: (b) The bottom-up beta, because the regression explains little of the variance

★ **Remember.** Note the direction: here bottom-up is lower, so it *raises* the valuation. Say so — disclosing that your choice helped the answer is what earns the mark.

SECTION D Terminal value, multiples and relative valuation

Terminal value usually carries 70–80% of enterprise value, so these are the highest-leverage assumptions in any model — and the ones examiners probe hardest.

27

Nominal growth

Long-run real GDP growth is 6.5% and long-run inflation 4.0%. What is nominal GDP growth, built the way the method requires?

- (a) 10.50% (b) 10.76%
(c) 2.50% (d) 26.00%

Working. Growth rates compound; they do not add.

$$g_{\text{nom}} = (1 + g_{\text{real}})(1 + i) - 1 = (1.065)(1.040) - 1 = \mathbf{10.76\%}$$

The additive shortcut gives 10.50%, which is 26 basis points too low. Over a perpetuity that error is not trivial.

Ans: (b) 10.76%

★ **Remember.** Terminal g must also stay below WACC, and a mature single-country firm should not out-grow nominal GDP forever.

28

Steady state

Terminal growth is 6.99% and terminal ROIC is 11.94%. What reinvestment rate does steady state require?

- (a) 41.4% (b) 58.6%
(c) 69.9% (d) 11.9%

Working. In steady state, growth is funded entirely by reinvesting a fraction of NOPAT at the return on new capital.

$$g = \text{RONIC} \times \text{reinvestment rate} \Rightarrow \text{reinvestment rate} = \frac{g}{\text{RONIC}}$$

$$= 0.0699 / 0.1194 = \mathbf{58.6\%}$$

So terminal FCFF = NOPAT $\times$ (1 – 0.586) = 41.4% of NOPAT. Forgetting to charge this reinvestment is the fastest way to overstate a terminal value.

Ans: (b) 58.6%

29

Justified multiple

Terminal NOPAT/EBITDA is 0.4902, terminal ROIC 11.94%, terminal growth 6.99% and ρ 11.39%. What EV/EBITDA do these fundamentals justify?

- (a) 19.4x (b) 4.65x
(c) 7.52x (d) 1.13x

Working.

$$\frac{EV}{EBITDA} = \frac{(NOPAT/EBITDA)(1 - g/ROIC)}{WACC - g}$$

Reinvestment rate = $0.0699/0.1194 = 0.5858$, so $1 - 0.5858 = 0.4142$.

Numerator = $0.4902 \times 0.4142 = 0.2031$.

Denominator = $0.1139 - 0.0699 = 0.0440$.

$EV/EBITDA = 0.2031/0.0440 = 4.65x$.

Ans: (b) 4.65x

★ **Remember.** The test is not whether your multiple equals the peer multiple. It is whether your multiple is arithmetically consistent with your own g , ROIC and WACC.

30

EV to invested capital

A company earns ROIC of 3.3% today, terminal growth is 6.99% and WACC 11.36%. What multiple of invested capital is justified?

- (a) 1.66x (b) 0.46x
(c) Negative — no positive multiple is justified (d) 2.02x

Working.

$$\frac{EV}{IC} = \frac{ROIC - g}{WACC - g} = \frac{0.033 - 0.0699}{0.1136 - 0.0699} = \frac{-0.0369}{0.0437} = -0.85x$$

The numerator is negative because the company earns *less* than the rate at which it is being asked to grow. Every rupee reinvested destroys value, so no positive multiple of invested capital is warranted at all.

This is the sharpest single statement of the Ambuja investment case.

Ans: (c) Negative — no positive multiple is justified

★ **Remember.** When $ROIC < g$, growth is value-destroying. When $ROIC = WACC$, growth is value-neutral. Only $ROIC > WACC$ makes growth worth having.

31

Reverse cross-check

A peer median exit multiple implies terminal growth of 10.33%, while nominal GDP growth is 10.76% and the company's realisation has compounded at 2.0% over ten years. What is the correct conclusion?

- (a) Adopt the peer multiple — the market knows best (b) The peer multiple is the implausible end of the range, and you should say so
(c) Raise your terminal growth to 10.33% (d) Relative valuation cannot be reconciled with DCF, so ignore it

Working. Solving a peer multiple back for the growth it implies is the reverse cross-check the method requires. Here it demands perpetual growth at roughly 96% of the whole economy's rate, from a company whose price has compounded at 2.0%.

A company growing faster than the economy forever eventually *becomes* the economy. Adopting that growth to justify the multiple would contradict your own model. Take a position, evidence it, and record the gap honestly.

Ans: (b) The peer multiple is the implausible end of the range, and you should say so

32

Terminal dominance

Two models value similar companies. Model A uses a five-year explicit horizon and terminal value is 79% of EV; Model B uses ten years and terminal value is 72%. What does this mainly reflect?

- (a) Model B is more optimistic
- (b) Horizon length, not optimism
- (c) Model A has a higher WACC
- (d) Model B has higher terminal growth

Working. Lengthening the explicit forecast moves cash flows out of the terminal perpetuity and into the modelled period. The *same* company modelled over ten years will always show a lower terminal share than over five.

The method's guidance: extend the explicit period until steady state is credible — particularly when $ROIC \gg WACC$ or growth is far above steady state. Report terminal value as a percentage of EV, and treat 80–90% as the level demanding scrutiny.

Ans: (b) Horizon length, not optimism

33

Choice of multiple

For which type of business is price-to-book the appropriate primary multiple?

- (a) FMCG and IT services
- (b) Banks, NBFCs, oil and gas, real estate
- (c) Any company with negative earnings
- (d) Companies with high operating leverage

Working. The selection rule follows where value actually sits on the financial statements.

Income-statement businesses → P/E or EV/EBITDA

Balance-sheet businesses → P/B, paired with ROE

For a bank, the balance sheet *is* the business, so book value is the meaningful scale variable. P/E is meaningless with negative earnings; P/B needs positive net worth.

Ans: (b) Banks, NBFCs, oil and gas, real estate

34

Valuing a bank

Why is an enterprise FCF model inappropriate for a bank, and what should be used instead?

- (a) Banks have no cash flows; use EV/EBITDA
- (b) Debt and working capital are the raw material, so value equity directly with DDM or FCFE at K_e
- (c) Banks are too volatile; use book value only
- (d) Use FCFF but with a higher WACC

Working. For an industrial firm, debt is financing. For a bank, deposits and borrowings are *inputs to production* — the thing it buys and sells. Separating operating from financing flows is therefore meaningless, and enterprise value has no clear content.

Value equity directly: DDM or FCFE, discounted at K_e

Reinvestment is driven by capital-adequacy requirements rather than by capex. The primary multiple is P/B paired with ROE, and residual income works well too.

Ans: (b) Value equity directly — DDM or FCFE at K_e

★ **Remember.** Stating this method switch prominently is itself graded — it demonstrates you understand risk consistency rather than just following a template.

35

Order of work

Where does relative valuation belong in the workflow, and why?

- (a) Before the DCF, to anchor the target price
- (b) After the DCF, because a multiple is an outcome of valuation rather than its cause
- (c) In place of the DCF when time is short
- (d) Only for loss-making companies

Working. A multiple is value divided by an accounting indicator — it is the *result* of a valuation, not an input to one. Anchoring a DCF to “the market multiple says so” reverses the logic and imports the market’s assumptions into what is meant to be an independent estimate.

The correct sequence: build the DCF, derive the multiple your own model implies, then place it beside the peer set and *explain* any gap through fundamentals — growth for P/E, ROE for P/B, returns for EV/IC.

Ans: (b) After the DCF, because a multiple is an outcome of valuation rather than its cause

SECTION E Applied: the cement valuations

Five questions using the actual figures from the Ambuja and UltraTech models, where the theory above has to survive contact with real accounts.

36

Basis error

Ambuja’s group market capitalisation is Rs 99,095 cr. Standalone EBITDA is Rs 2,930 cr; consolidated EBITDA is Rs 6,559 cr; non-controlling interest is Rs 12,499 cr and the group is roughly net-cash. Which EV/EBITDA is correct, and why?

- (a) 34.0x, on standalone EBITDA
- (b) 17.0x, on consolidated EBITDA with NCI added to EV
- (c) 15.1x, on consolidated EBITDA without NCI
- (d) Either is acceptable if disclosed

Working. Market capitalisation prices the *whole group*, including the 50.05% stake in the separately listed ACC. Standalone EBITDA excludes ACC entirely. Dividing one by the other is a category error.

Wrong: $99,676 / 2,930 = 34.0x$

Right: $(99,095 + 12,499 - 93) / 6,559 = 111,501 / 6,559 = 17.0x$

Non-controlling interest is added to enterprise value because consolidated EBITDA includes 100% of the partly owned subsidiaries that produce it. Numerator and denominator must cover the same set of assets.

Ans: (b) 17.0x, on consolidated EBITDA with NCI added to EV

★ **Remember.** Whenever a multiple looks absurd, check the basis before you check the market.

37

The bridge

Ambuja's consolidated cash flows include 100% of its partly owned subsidiaries. Non-controlling interest is Rs 12,499 cr and there are 247.18 cr shares. What happens if the NCI is not deducted in the bridge?

- (a) Nothing — NCI is a book entry (b) Value per share is overstated by about Rs 51
(c) Value per share is understated by about Rs 51 (d) Only the WACC changes

Working. If the cash flows being discounted belong 100% to the group, but only part of the group belongs to Ambuja's shareholders, the minority's share must be removed before dividing by Ambuja's share count.

$$12,499/247.18 = \text{Rs } 50.57 \text{ per share}$$

Omitting it hands the parent's shareholders value that belongs to the minorities — liability consistency again, in its most expensive form.

Ans: (b) Value per share is overstated by about Rs 51

38

After-tax returns

A workbook computes historical ROIC as EBIT of Rs 11,761 cr over invested capital of Rs 81,733 cr, but the cell meant to apply tax references an empty cell. Statutory tax is 25.168%. What is the reported ROIC, and what is the correct one?

- (a) Reported 14.39%, correct 10.77% (b) Reported 10.77%, correct 14.39%
(c) Both 14.39% (d) Reported 14.39%, correct 12.50%

Working. With the tax reference broken, NOPAT collapses to EBIT and the series becomes a *pre-tax* return.

$$\text{Reported: } 11,761/81,733 = 14.39\% \quad (\text{pre-tax})$$

$$\text{Correct: } 11,761 \times (1 - 0.25168)/81,733 = 10.77\%$$

The distinction matters enormously here: 14.39% sits comfortably above an 11.4% cost of capital, while 10.77% sits *below* it. The same company looks value-creating or value-destroying depending on one cell reference.

Ans: (a) Reported 14.39%, correct 10.77%

★ **Remember.** ROIC is always after tax. A pre-tax return compared with a WACC is comparing unlike with unlike.

39

Normalising a tax credit

Ambuja reports a net tax *credit* of Rs 2,338 cr, so reported profit exceeds pre-tax profit and the effective rate is minus 71%. How should the valuation treat it?

- (a) Use the reported effective rate of -71% (b) Use the statutory rate throughout and decompose the credit by its economics
(c) Ignore tax entirely for that year (d) Average the reported rate over five years

Working. A negative effective rate is unusable as a forward-looking assumption. Tax EBIT at the statutory 25.168% throughout, then treat each component of the credit on its economics, not its accounting:

- Provision reversals (Rs 1,251 cr) — non-cash, prior period. **Excluded:** the credit was never in the cash flows.
- Tax paid under protest (Rs 614 cr) — real cash already handed over, recoverable. **Added** as a non-operating asset.
- Deferred tax liability (Rs 3,466 cr) — a real claim. **Deducted** once in the bridge.

Net effect: about **minus Rs 11.54 per share**. A Rs 2,338 cr headline gain is worth negative value.

Ans: (b) Use the statutory rate throughout and decompose the credit by its economics

40

EVA and enterprise value

A company's EVA is negative in nine of ten forecast years. What must be true of its enterprise value?

- | | |
|-------------------------------------|--|
| (a) It must be negative | (b) It must sit below invested capital |
| (c) It must exceed invested capital | (d) Nothing follows |

Working.

$$EV = IC_0 + PV(\text{future EVA})$$

If EVA is persistently negative, the present value of future EVA is negative, so EV falls below opening invested capital. That is not an error — it is exactly what the theory predicts when a company earns less on capital than capital costs.

EV can still be comfortably positive; it is simply worth less than the capital sunk into it. Equivalently, $EV/IC < 1$, which the identity in Q30 also produces.

Ans: (b) It must sit below invested capital

★ **Remember.** Negative EVA with a positive EV is perfectly coherent. Do not "fix" it.

SECTION F Guest lecture — M&A, synergy and exchange ratios

Five questions on the transaction module. Value each entity standalone first, then value only the incremental synergy cash flows, at a rate that matches their risk.

41

Definition of synergy

Acquirer A is worth Rs 800 cr standalone and target B Rs 300 cr standalone. The combined entity is valued at Rs 1,250 cr. What is the synergy value?

- (a) Rs 1,250 cr (b) Rs 450 cr
(c) Rs 150 cr (d) Rs 950 cr

Working. Synergy is the value the combination creates over and above what the two businesses are already worth apart.

$$\text{Synergy} = V_{AB} - (V_A + V_B) = 1,250 - (800 + 300) = \text{Rs } 150 \text{ cr}$$

Note what this requires: both standalone valuations must be built *first* and on a consistent basis. Synergy is a residual, so any error in either standalone figure lands entirely in the synergy number.

Ans: (c) Rs 150 cr

42

Net gain to the acquirer

On the same facts, A pays Rs 380 cr for B. What is the net gain to A's shareholders?

- (a) Rs 150 cr (b) Rs 80 cr
(c) Rs 70 cr (d) Rs 380 cr

Working. The premium is what the acquirer hands to the target's shareholders; the net gain is what is left of the synergy after paying it.

$$\text{Premium} = 380 - 300 = \text{Rs } 80 \text{ cr}$$

$$\text{Net gain to A} = \text{Synergy} - \text{Premium} = 150 - 80 = \text{Rs } 70 \text{ cr}$$

The transaction creates Rs 150 cr of value, of which A's shareholders keep Rs 70 cr and B's shareholders capture Rs 80 cr. If the premium had exceeded Rs 150 cr, A would be transferring value to B despite a genuinely synergistic deal.

Ans: (c) Rs 70 cr

★ **Remember.** A good deal at a bad price is a bad deal. Always report the net gain after the premium, not the synergy alone.

43

Exchange ratio

A share-swap merger values the target at Rs 480 per share and the acquirer at Rs 1,600 per share. What is the exchange ratio?

- (a) 0.30 acquirer shares per target (b) 3.33 acquirer shares per target
(c) 0.48 acquirer shares per target (d) 1.60 acquirer shares per target

Working. The exchange ratio equates the value given to the value received, per share.

$$ER = \frac{\text{value per target share}}{\text{value per acquirer share}} = \frac{480}{1,600} = \mathbf{0.30}$$

So each target shareholder receives 0.30 acquirer shares. If the target has 100 cr shares, the acquirer issues 30 cr new shares.

The ratio should be set on *intrinsic* values, including the target's share of synergy that the acquirer is willing to pay away — not on traded prices alone, which already contain deal speculation.

Ans: (a) 0.30 acquirer shares per target share

44

EPS accretion

A merger is EPS-accretive in year one. What does this tell you about value creation?

- (a) The deal creates value (b) Nothing reliable — accretion depends on the relative P/E ratios
(c) The deal destroys value (d) The synergies must be real

Working. A company with a *higher* P/E acquiring one with a *lower* P/E will report EPS accretion arithmetically, with no synergy whatever, simply because it buys more earnings per share issued. The reverse produces dilution even when the deal is excellent.

Accretion is an artefact of relative P/E, not evidence of economic value

Value creation is tested by discounting incremental synergy cash flows and comparing them with the premium paid — Q41 and Q42, not the EPS line.

Ans: (b) Nothing reliable — accretion depends on the relative P/E ratios

★ **Remember.** Never substitute EPS accretion for economic value creation. This is the single most tested point in the transaction module.

45

Valuing synergies

Which set of cash flows may be included when valuing synergy, and at what rate?

- | | |
|--|---|
| (a) All the target's cash flows, at the acquirer's WACC | (b) Only incremental, timing-specific flows, at a risk-matched rate |
| (c) The combined entity's total cash flows, at the target's WACC | (d) Cost savings only, at the risk-free rate |

Working. Three disciplines apply.

Incremental. Only cash flows that would not exist without the combination. The target's standalone flows are already in V_B ; counting them again double-counts.

Timing-specific. Synergies arrive on a schedule and cost money to achieve. Implementation costs and their timing must be explicit.

Risk-matched. Cost synergies are usually more certain than revenue synergies, so they do not deserve the same discount rate. Revenue synergies should carry a higher rate, or a probability weighting.

Treat revenue, cost, tax and financing effects separately, and sensitise the assumptions that drive the premium decision.

Ans: (b) Only incremental, timing-specific cash flows, discounted at a rate matching their own risk

★ **Remember.** Revenue synergies are the ones that most often fail to arrive. Examiners expect you to say so.

— end of question bank —

45 questions | 40 core + 5 guest lecture